Covid-19 Tracker: A data visualization tool for time series data of pandemic in India

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***Abstract*—Covid-19 first originated in Wuhan, China, became a global pandemic and spread to many countries due to its high transmissibility. Digital technology could prove to be of great use in pandemic management with many strategies like screening for infection, tracking, contact tracing, and many more. Covid-19 tracker is one such application of digital technology. This helps in visualizing the pattern of public health through graphs, search results, and tables that can be easily understood by the user. It is built using HTML and JavaScript. Covid-19 Tracker uses data from certain authentic sources and helps to visualize the spread of Covid-19 throughout India. The main features incorporated in our dashboard are: total cases reported, graphs for daily trends, search functionality, state-wise comparative evaluations, and hotspot distribution map of the entire country. This dashboard provides a variety of user involvement possibilities and extracts useful data in a simple and easy-to-understand way.**

***Index Terms*—Covid-19, Dashboard, Data visualization**

I. Introduction

Covid-19 is a global pandemic caused by the novel SARS-CoV-2 coronavirus. The first known case was detected in Wuhan, China, in December 2019. The disease has since spread around the world, resulting in a pandemic. Containment and moderation have been the mainstay of pandemic management due to its high transmissibility, a lack of effective antiviral medication or vaccine, and a case fatality rate of greater than 1%. Pandemic strategy can be aided by digital health technology in ways that are impossible to achieve manually. The necessity to keep track of Covid-19 has sparked the development of data dashboards that visually depict illness burden and pattern of public health [2]. Every country has gathered data related to Covid-19 like total recovered cases, confirmed cases, deaths, etc. But this raw form of data makes it difficult for people to understand and visualize the patterns of disease's growth. Dashboards analyze and present raw data as visualizations, graphs, and text, with a variety of user involvement possibilities, and make it possible to extract useful data and present it in a simple and easy-to-understand way. By studying these trends, it would be very helpful for governments in identifying patterns. The patterns discovered from these visualizations will in turn help policymakers and epidemiologists to suggest preventive measures. This dashboard is implemented by use of HTML and Javascript. This paper will focus on web dashboard's development with Covid-19 data, for the country, India. The features incorporated in our dashboard are total cases reported, graphs for daily trends, search functionality, state-wise comparative evaluations, and hotspot distribution map of the entire country. This dashboard provides user interaction and is easy to use and understand.

II. Related Work

Since Covid-19 has been declared as a pandemic many different types of research have been done related to this. The research in this field using digital technology has been proved very useful in pandemic management. The researches have come up with solutions like data driven visualization [3] [4] [5], prediction techniques [6], visual analytics of Covid-19 trends [7] [8], and many more. Until now, many visualizers and dashboards have been created. The dashboards present the data in an easily understandable format and can be hosted on any web server. Many institutions like World Health Organization(WHO) [9] and John Hopkins Corona Virus Research Centre [10] have developed dashboards with global real-time data. They provide general information regarding the worldwide cases and at the country level. The most common attributes that the majority of dashboards use are confirmed cases, recovered cases, and deaths. The majority of these solutions use maps, line, or bar charts to represent the patterns, as well as a table to display the values. All of these platforms give trend analysis, but none of them include thorough visualizations or diverse visualization formats for each country, especially India.

Our dashboard provides in-depth data visualization of India at a state level. And has different visualization formats which include a date range daily trends graph, statewise comparative chart, data table, and a bubble map which will give the spatial

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information regarding the intensity of the cases in every state. By using this dashboard we can understand the evolution of Covid-19 in India.

III. Data Set and Data Processing

Data is of JSON format and is extracted from “api.covid19india.org/data.json” which provides up-to-date data of cases for India. This data has data up to t-3 days i.e. till 3 days before the current date. The dataset consists of case time-series data and statewise data of the country India. The case time series data contains daily confirmed, daily deceased, daily recovered, date, date in y-m-d format, total confirmed, total recovered and total deceased. And the state-wise data contains active, confirmed, deaths, delta confirmed,delta recovered, last updated time, migrated other, recovered, state, state code, and state notes. The data is fetched from “api.covid19india.org/data.json” using $.get() method which loads the data from the server using an HTTP Get request. And then, the selected and useful data to work with is stored in arrays which makes it easier to work with data and use it according to the requirements.

IV. Features and Implementation

The tracker provides visualization of the number of cases and deaths on a platform where the users can see different trends across the whole country and also according to the states. Real-time data is used and visualized in the dashboard. The main objective of this tracker is to represent cases, deaths, and recoveries through data visualization. It uses HTML and Javascript for its application. There are total 6 features in this dashboard which are as follows:

- Tags: There are four tags named Confirmed, Active, Recovered and Deaths which represent the total respective number below them. This is done by getting references to the ids from HTML and appending the values fetched from the data source to those ids.
- Daily trends graph with date range selector: It shows the cumulative daily trend of deaths, confirmed and recovered cases from 30^{th} January 2020 to till date and this is the default graph which is shown in the dashboard as depicted in Fig. 2. It has a date selector which will plot a graph between the selected starting and an ending date. It has been done using the date range picker library of JavaScript and the graph is made using the chart.js library of JavaScript.
- Search bar: It shows data of that particular state which the user searches for, under the four tags named Confirmed, Active, Recovered, and Deaths. This is done similarly as tags but while showing specific results for the query that has been searched.
- State-wise graph: This features a detailed state-wise comparative chart of total confirmed cases, recovered cases, and deaths. The graph is made using the chart.js library of JavaScript. It also has toggle buttons, ‘line’ and ‘bar’ where it can change the chart type when clicked upon that particular button. This helps the user to understand the

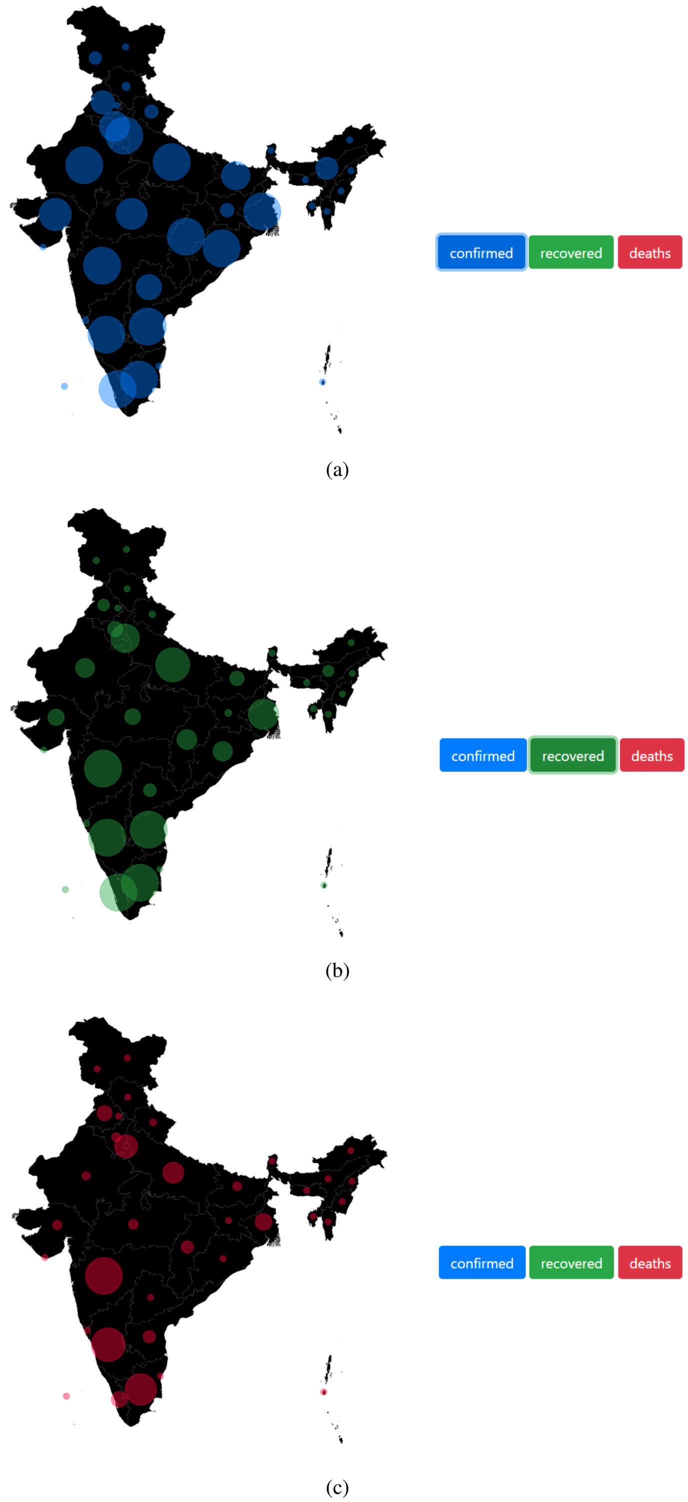

Fig. 1. (a) Confirmed hotspot map (b) Recovered hotspot map (c) Deaths hotspot map

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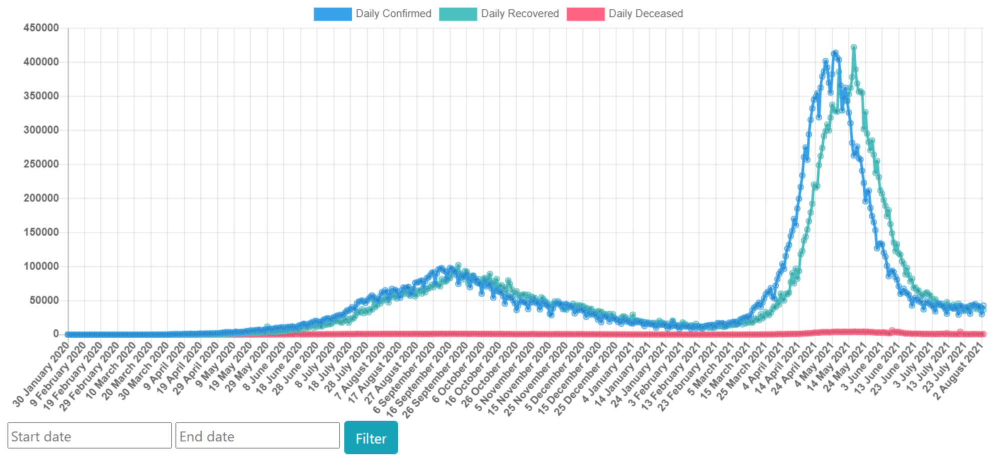
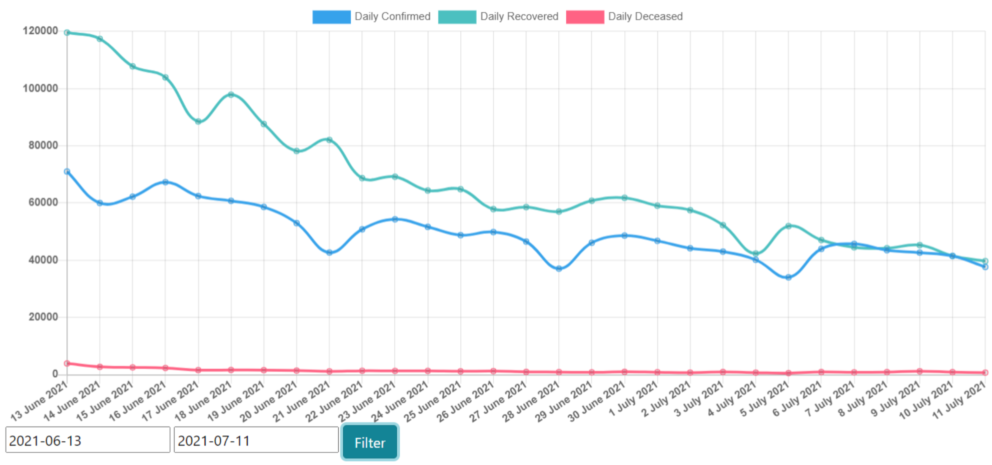


(a)



(b)

Fig. 2. (a) Before entering date range (b) After entering date range

comparison more clearly with the help of two different views as shown in Fig. 3.

- State-wise table: It is the tabular representation of data that is fetched from the data source. It has 6 attributes and 36 tuples. A view of part of this tabular representation is shown in Fig. 4.
  Attributes: State name, state code, confirmed cases, active cases, recovered cases and deaths.
  Tuples: All state names and a of total of all data.
- Maps: It's a hotspot map that depicts case hotspots. There are three buttons: confirmed, recovered, and deaths. The map depicts the hotspot of specific cases of the button that has been clicked. It also shows the number of cases and state name when the cursor hovers upon any bubble. Fig. 1 represents the view of these hotspots. These maps represent spatial information about the total number of confirmed cases, recovered cases, and deaths of different states [11]. The radius of the hotspot circle is set relatively by finding the highest cases in each dataset of confirmed, recovered and deaths, and then the radius will be calculated by (cases/highest case) $\times 100$.

V. Conclusion and Future Scope

Millions have died as a result of the Covid-19 pandemic, which has prompted countries to take appropriate measures to prevent the spread. Digital technology has shown to be quite useful in the management of pandemics in recent years. Dashboards were intended to meet this demand. Interactive dashboards allows us to display data in a way that is easier to decipher and comprehend for the human brain. We created an interactive dashboard that allows you to load and view data, such as trends, comparative charts, tables, and maps. It gives thorough information in a variety of formats to help people understand and learn about the evolution of Covid-19 in India. In this paper, many different formats of data visualization are demonstrated. The two crests evident in the daily trends graph clearly show the first and second waves of Covid-19 in India. As we can observe in Fig. 2, the first wave peaked in September 2020, while the second wave peaked in early May 2020. We can also see that the second wave was twice as lethal

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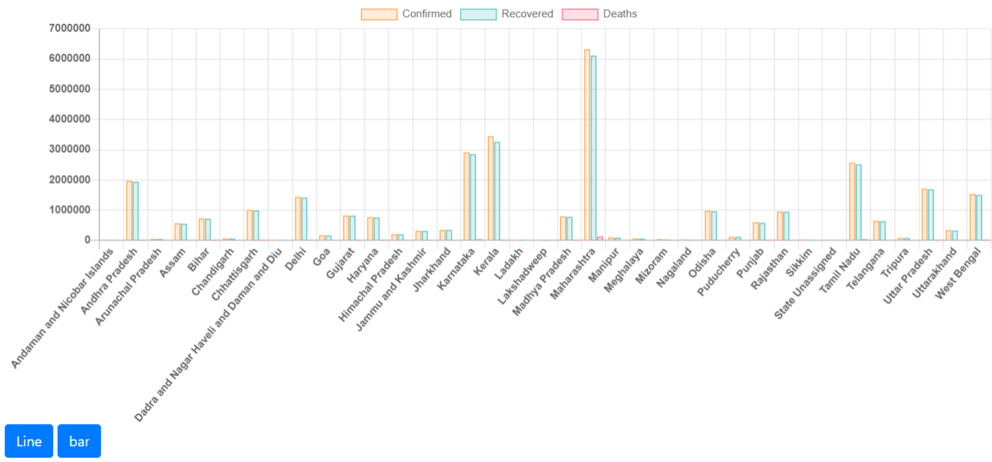
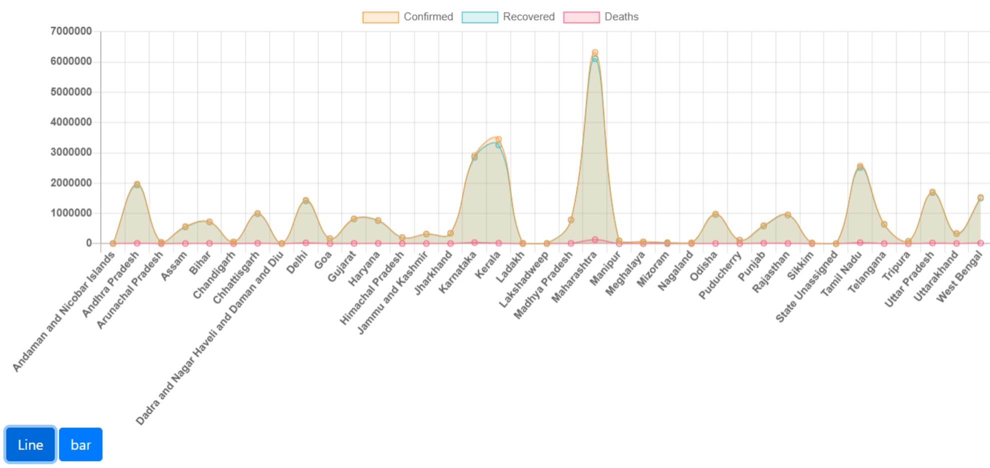


(a)



(b)

Fig. 3. Toggle: (a) Bar graph (b) Line graph

CO-VID 19 STATE-WISE

State	State Code	Confirmed	Active	Recovered	Deaths
Total	TT	31767985	404262	30925354	425789
Andaman and Nicobar Islands	AN	7540	6	7405	129
Andhra Pradesh	AP	1971554	20170	1937956	13428
Arunachal Pradesh	AR	48884	3352	45298	234
Assam	AS	569439	11093	551692	5307
Bihar	BR	724977	383	714949	9644
Chandigarh	CH	61965	34	61120	811
Chhattisgarh	CT	1002600	1881	987189	13530
Dadra and Nagar Haveli and Daman and Diu	DN	10654	17	10602	4
Delhi	DL	1436451	519	1410874	25058
Goa	GA	171396	999	167245	3152
Gujarat	GJ	824939	226	814637	10076
Haryana	HR	769982	708	759633	9641

Fig. 4. State-wise table

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as the first. We can deduce from the comparison state-by-state graph that Maharashtra is the worst-hit state of all. The bubble map or the hotspot map gives the spatial information regarding the number of cases and their intensity on a state which is represented by dynamic radii. This could be useful for the community in finding and avoiding pandemic hotspots. These insights could aid policymakers and the government in taking necessary action.

We would like to update the website's appearance soon. We'd also like to include other visualizations, such as district-specific data and its impact on virus containment, as well as micro-level data concerning patient characteristics such as age, gender, and so on, to provide a complete picture of the Covid-19 virus's spread in India.

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